



Jaguar launches InMotion

Jaguar Land Rover has launched a startup studio which will design mobility apps and services.
<http://dgit.in/JInmotion>



Kindle Oasis launched

Amazon's upcoming premium Kindle priorly leaked with a major change in the design.
<http://dgit.in/Amoasis>

devices were mostly used for the input of graphical data into computers, much like modern day digital pens. In fact, Apple had its own touch based tablet interface way back in the late 1970s. The Apple Graphics, which was designed by Summagraphics and also marketed as BitPad, was initially sold as an accessory with the Apple II Personal Computer. Actually, this device wasn't a standalone tablet in the likes of today's tablet PCs. Instead this was an interface for graphical input for the Apple II. The interesting fact is, this predated the Apple II's mouse by almost 6 years.

A fast forward to 1993 brings us to one of the most interesting devices on the touch front - The Eo Personal Communicator. It was a notebook sized fullscreen touch-screen device that ran PenPoint OS, one of the first operating systems to be designed targeted towards the use of styluses. Unfortunately, the company was shut down after it failed to meet revenue targets set by AT&T that had recently acquired it.

As the components became smaller and more powerful, the desire for smaller compact devices gave rise to the Personal Desktop Assistants, Or PDAs. As the touchscreens went smaller, the mode of input had to be redesigned as a full keyboard would no longer fit on the handheld PDAs. This requirement gave rise to new and innovative methods of input on these devices. One such method was used by Palm in its PalmPilot range of products back in 1997 and it was called Graffiti. Each device had a touch sensitive area to detect stylized characters drawn with the included stylus. There was also a tutorial game included which taught players how to draw stylized versions of characters falling on the screen.



The PalmPilot being used with the Graffiti gesture based input system

The feature phones arrive

PDAs were somewhat expensive for general usage and with components further reducing in size, there was an opportunity for feature phones to rise and become popular devices all around the world. This also gave rise to one of the most used forms of communication worldwide - texting. Since phones had to use physical buttons, it became a challenge to include an entire keyboard with full features at the size of the keypad. There were many suggested methods and the one that stuck was the one that is still seen on keypad phones. A single button press indicated the first letter for that key, two presses for the second letter and so on. As phone screens grew larger, this system grew



The BlackBerry Torch keyboard

more sophisticated with the inclusion of dictionaries on phones as well as the T9 system that began predicting words based on the letters being typed.

One of the unique innovations on this front was from BlackBerry. They included a full featured keyboard on the phone with buttons large enough to be used with the thumbs. This keyboard is still very popular with frequent texters and is often included as a slide-out under the original phone.

Touch 2.0

While we already know that touchscreens and touch screen devices weren't really invented quite recently, the innovation that would be the next big breakthrough was still in the making even a couple of decades ago. A pioneer gesture recognition tech company, FingerWorks, released a range of multi-touch products

VIRTUAL REALITY

We are quite aware of Virtual reality being the next big computing interface at present. With its completely different approach to interaction, it is evident that there are new methods of input. The HTC Vive uses controllers that can be detected by its lighthouse sensors and onboard cameras to mimic hands in the virtual reality. There are other alternatives that can directly sense your hands as well. A combination of these sensors along with modification in the platform could very well make the future input devices completely virtual as well.

in 1998 including the iGesturePad and TouchStream keyboard. This company was acquired by Apple in 2005.

Less than two years later, in 2007 Apple released the iPhone with the most innovative touchscreen technology anyone had ever seen, especially on a handheld device. The phone interface was completely touch based, with every part of the screen having a functionality on every interface in the OS, including the complete virtual keyboard

The widespread usage of the stylus changed when Apple implemented a multi-touch capacitive touchscreen on the iPhone, which was highly sensitive and could detect several contacts at once to allow gestures like "pinch" and "zoom" in the OS. The entire OS was designed to take advantage of this touchscreen with completely new ways to use standard applications like the calculator, camera, games etc. Before we realised, gestures like swipe to scroll, inertial scrolling, long-press, press

and drag became commonplace implementations on any and every touchscreen device and mechanisms like pattern locks, tap to focus in camera, gesture based commands became commonplace. Apple changed the phone standards overnight and stylus based touchscreens almost immediately became obsolete. This widespread adoption is still evident in the regularly ongoing copyright infringement lawsuits between



The iPhone, First generation